CODEFORCES
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PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

G. Sorting Again??

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Suppose you are given a task to rank the students. You have gotten the marks and ID of the students. Now your task is to rank the students based on their marks using a sorting algorithm. If two or more students get the same mark, then students with the lower ID will get prioritized. See the input and output for a better understanding.

However, **you have to keep in mind that your sorting algorithms perform the minimum number of swapping operations.**

Input

The first line contains an integer T $(1 \leq T \leq 100)$ denoting the number of test cases.

For each test case:

The first line contains an integer N $(1 \leq N \leq 1000)$.

The second line contains N distinct integers S_i $(1 \leq S_i \leq 1000)$, where S_i denotes the student ID.

The third line contains N integers S_m $(1 \leq S_m \leq 1000)$, where S_m denotes the obtained mark of the corresponding student.

Note: It is guaranteed that all student IDs are unique.

Output

For each test case, the output must contain a number X which denotes the number of minimum swaps. The rest of the N lines will contain the Student ID and obtained marks sorted based on the instruction above. See the sample output for a better understanding.

Important Note: Since you are asked to minimize the number of swaps, if your number of swaps doesn't match with the judge's answer, your solution will be considered incorrect.

TO CHANGE: Look at the first testcase of the sample input. It can be shown that this can be sorted with only 4 swaps. It can also be shown that it is not possible to sort this in less than 4 swaps.

Example

input Copy

```
2
7
7 4 9 3 2 5 1
40 50 50 20 10 10 10
3
1 2 3
40 59 10
```

output Copy

```
Minimum swaps: 4
ID: 4 Mark: 50
ID: 9 Mark: 50
```

CSE221 Assignment 01 Fall 2026
Contest is running
9 days
Virtual Participation

→ Submit?

Language: Java 21 64bit
Choose file: Choose File No file chosen
Success!
CLOUDFLARE
Privacy • Help
Submit

```
ID: 7 Mark: 40
ID: 3 Mark: 20
ID: 1 Mark: 10
ID: 2 Mark: 10
ID: 5 Mark: 10
Minimum swaps: 1
ID: 2 Mark: 59
ID: 1 Mark: 40
ID: 3 Mark: 10
```

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